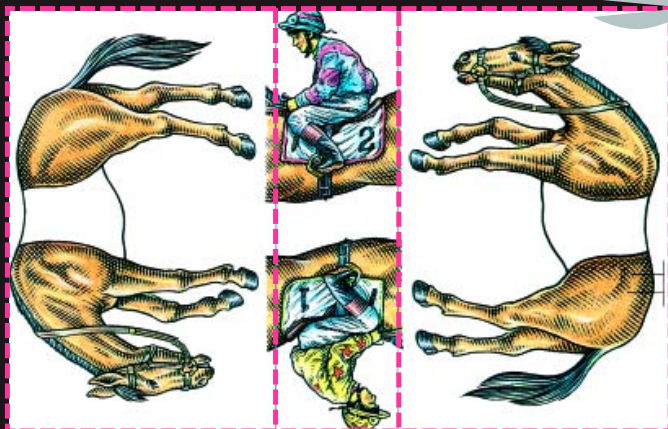
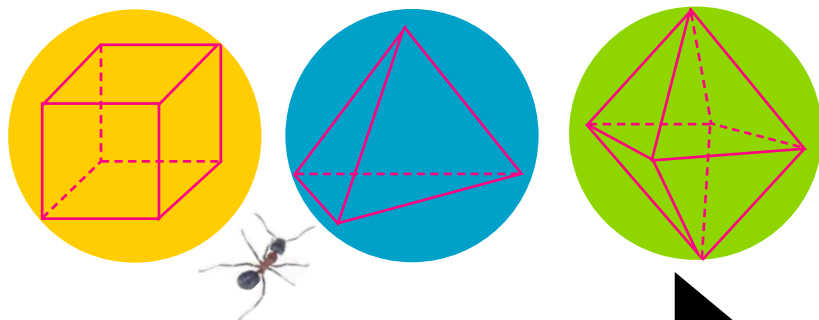




8 Horses and riders

Trace or photocopy this drawing, then cut along the dotted lines to make three pieces. How can you arrange the pieces so that each rider is correctly riding a horse, without folding or cutting any of the pieces?



9

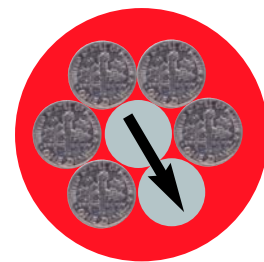
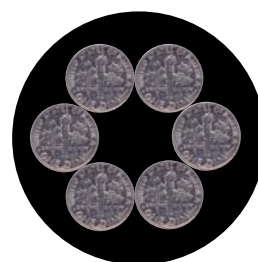
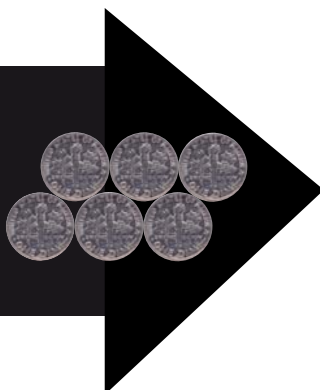
The exploring ant

Imagine an ant walking around these shapes. Can it walk along all the edges of each shape without retracing its path? To find out, try drawing each shape as one line, without lifting your pen. Can you work out the rule that determines which shapes the ant can walk around?

10

Sliding coins

Arrange six coins in a parallelogram. How can you change the shape into a circle by moving only 3 coins? You can't nudge coins aside, and each coin you move must end up touching two others.



The move above is forbidden because you're not allowed to nudge coins out of the way.

11

Coloured cubes

A cube has been painted so that each of its six faces is a different colour. The three pictures here show it in different positions. Which colour is face down in the third picture? *Clue: try turning the cube around in your mind.*



12

Through the paper

By cutting along a clever pattern, it's possible to make a hole in a postcard-sized piece of paper that a person can step through. Can you work out the pattern?

